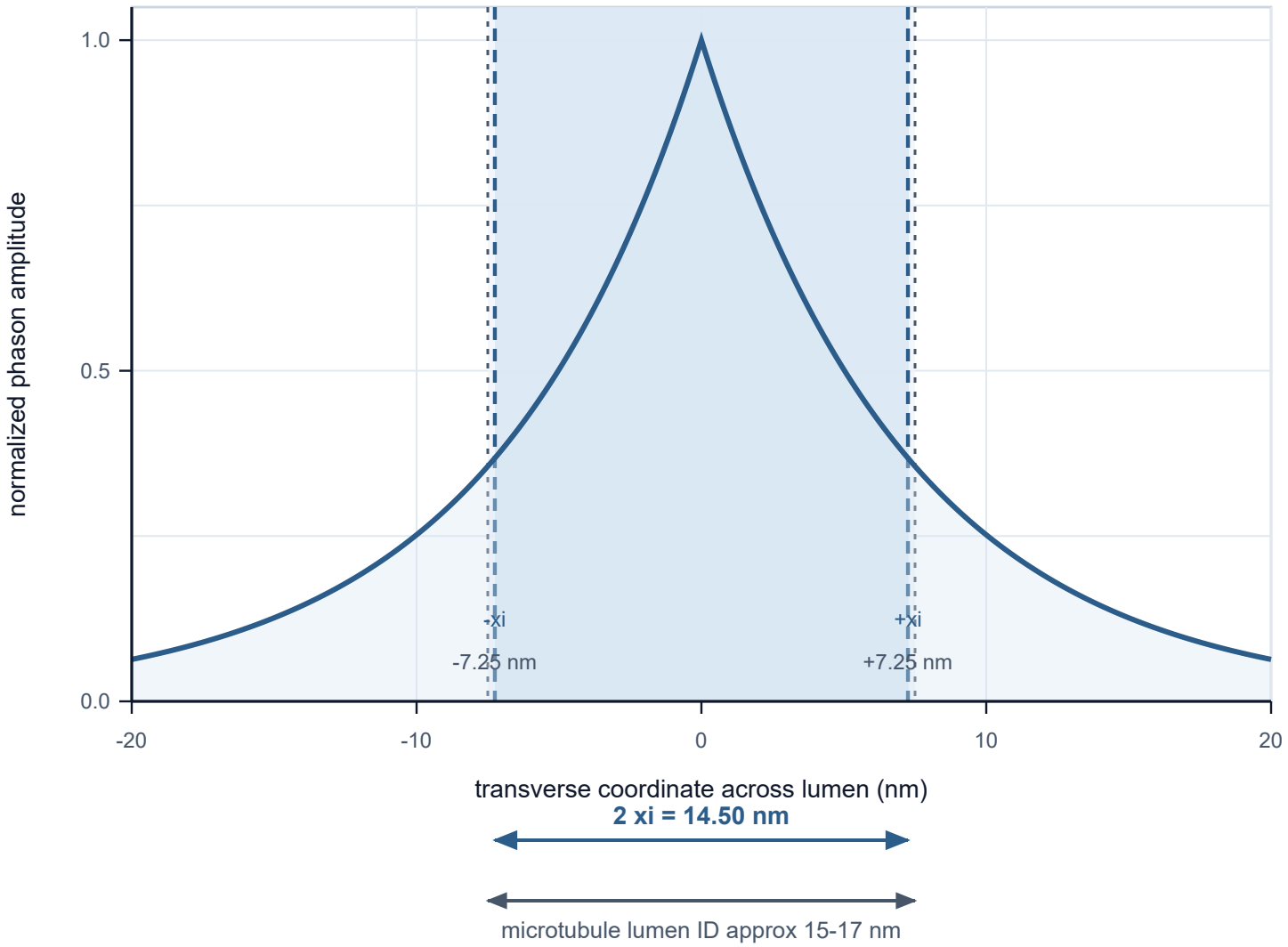
Figure V1.7.3

Phason healing length and lumen scale

Localized transverse profile: xi = a0/alpha; 2 xi aligns with the microtubule lumen scale.



Source quantities

a0 = 0.05291772 nm
alpha inverse = 137.036
xi = a0 / alpha = 7.25 nm
2 xi = 14.50 nm
K_perp / K_parallel
equals phi power -18
= 1.731e-04

xi: Tier 1 Bohr scaling
lumen match: Tier 3 correlation

Scale comparison

polaron diameter 14.50 nm

lumen diameter approx 15-17 nm

Interpretation: quantitative boundary-condition correlation only; not a first-principles derivation of the lumen geometry.